The Open Source Firefighter

Inspector’s Review Draft — Installation Requirements, All Enclosure Classes

v1.0 · August 2026 · Installation requirements in code-adoption language · Derived from the Hardware Annex and Main Specification *Document ID:* ***OSF-3.3*** *· one document of the Open Source Firefighter specification family.*

Andrew Morton · Palmer Lake, Colorado

This work is dedicated to Emily Barker, Nick Hutcherson, and Sydney Watson of the Rifle Helitack crew, killed in a burnover during initial attack on the Knowles Fire, western Colorado, June 27, 2026 — and to the two crewmates who survived it, and to all wildland firefighters who stand between fire and the rest of us.

They were the first to arrive. This system exists so that the ground they arrive on is already fighting back.

Reviewer’s note: This draft renders the installation-facing and inspection-facing requirements of the OSF hardware specification in code-format numbered paragraphs for plan-check style review. Mark it up the way you would mark up a submittal. Comments on anything are welcome — enclosure, mounting, access, wiring, plumbing, labeling, documentation, and especially anything that would generate a correction notice in the field. Network protocol and coordination behavior are out of scope for this review and are summarized only where an inspector would encounter their physical evidence.

Chapter 1 — Scope and Application

1.1 This document covers the installation, mounting, power, plumbing, marking, and documentation requirements for Open Source Firefighter (OSF) units in three enclosure classes: Class 1 (Residential), Class 2 (Government/Commercial), and Class 3 (Tactical/Transportable).

1.2 All three classes shall contain identical control, radio, and coordination internals. Class distinctions shall be limited to packaging, connectors, power sourcing, and environmental rating.

1.3 Installations shall comply with locally adopted building, plumbing, and electrical codes. Nothing in this document shall be construed to waive local permitting requirements.

1.4 The unit is a supplemental exterior wetting system for ember defense. It is not a fire alarm system (NFPA 72 / UL 864 class), not a fire suppression system as defined by the sprinkler codes (NFPA 13, 13D, 13R), and not a life-safety notification system, and shall not be represented as any of these. It is supplemental to, and replaces no requirement of, the wildland-urban interface provisions a jurisdiction has adopted — commonly the International Wildland-Urban Interface Code, NFPA 1140, or NFPA 1144.

Chapter 2 — Definitions

2.1 Zone. A valved water discharge circuit serving a defined coverage area.

2.2 Zone Manifold. The basic intake manifold header carrying the standalone zone valves, inlet gallery, check valve, drain, and pressure transducer, each valve individually mounted and individually replaceable.

2.3 Coordination Domain. The set of units sharing radio coordination. Physical evidence at inspection is limited to antennas and their penetrations.

2.4 Sequencing. Default operation in which one zone is open at a time on a rotating schedule.

2.5 Assembled Penetration. An enclosure penetration formed by a sealed component’s own mounting stub rather than by a discrete bulkhead fitting.

Chapter 3 — General Requirements (All Classes)

3.1 Enclosure

3.1.1 Wall-mounted enclosures shall be rated NEMA 4X or better.

3.1.2 Enclosures shall be hinged and lockable.

3.1.3 Where a sealed component forms its own enclosure penetration, the penetration shall be evaluated against the

enclosure rating as assembled.

3.1.4 All penetrations shall be gasketed or otherwise sealed consistent with the enclosure rating.

3.1.5 Batteries shall be mounted on a dedicated shelf, positioned per 3.1.6 through 3.1.9.

3.1.6 Interior hydraulic/electrical cohabitation shall follow established sprinkler valve-enclosure and irrigation-controller practice: control electronics shall be mounted above the wet works, and no electrical component shall be mounted beneath a mechanical plumbing joint.

3.1.7 All conductors descending toward the hydraulic zone shall include drip loops.

3.1.8 A drainage path shall be provided from the enclosure low point to the exterior.

3.1.9 Batteries shall not be placed below the hydraulic zone without a drained tray.

3.1.10 Sectioned two-compartment enclosures, or dual-enclosure formats separating electrical and hydraulic equipment, are equally compliant and are recommended for first-time builds.

3.2 Electrical

3.2.1 The internal bus shall be 24 VDC.

3.2.2 Power supply and battery charging shall be provided by standard DIN-rail power modules. Charging circuitry integrated onto control hardware shall not be permitted.

3.2.3 A low-voltage disconnect shall be provided to prevent battery deep discharge.

3.2.4 Batteries shall be commodity form factor, field-replaceable without special tools, and replaceable without disturbing control wiring.

3.2.5 Electrical installation shall comply with the locally adopted edition of the National Electrical Code or local equivalent.

3.2.6 Standby capability shall be a minimum of 72 hours for controller, radio, and positioning functions.

3.3 Plumbing

3.3.1 All wetted connections shall be NPT. Garden-hose-thread adapters are permitted as accessories on Class 1 units.

3.3.2 A spring check valve and a manual isolation valve shall be provided on the inlet gallery.

3.3.3 Backflow prevention shall comply with local water district requirements.

3.3.4 A manually valved drain-down port shall be provided at the manifold low point.

3.3.5 In freeze climates, the installation shall support a winterization procedure consisting of: all zone valves opened, supply isolated, manifold drained, system depressurized. The procedure shall be firmware-assisted and physically completed at the drain valve.

3.3.6 The zone manifold shall be a basic header feeding standalone valves, each valve individually removable without disturbing the others or the header.

3.4 Controls and Operator Interface

3.4.1 A local operator interface shall be provided on the unit, by which an operator physically present can, at minimum: initiate protection manually; disable or lock out automatic operation; command each zone independently; and signal distress to the coordination domain by a deliberate action of approximately three to five seconds' duration. These functions are controller inputs and are not emergency-stop devices; no safety category is claimed for them. No software configuration shall disable an operator control (Charter Clause 7). The hardware means of removing power is the disconnect; loss of power drives the auto-return valves closed.

3.4.1.1 The interface shall be operable by an operator wearing structural firefighting gloves, in darkness, without instruction, without a personal device, and without any network or account. It shall present its state without requiring the operator to interpret a display. *The reference implementation is discrete panel-mounted pushbuttons with indicator lamps; any implementation meeting the requirements of 3.4.1 and 3.4.1.1 is compliant.*

3.4.2 The unit shall indicate, at minimum, the following states in a manner readable at a glance without a display or a connected device: power, radio link, readiness, fault, active operation, and distress. *Panel indicator lamps are the reference implementation.*

3.4.3 The unit shall annunciate status audibly in plain language, including specific fault conditions and a periodic

maintenance reminder.

3.4.4 Configuration shall be by local direct connection only. No cloud service, remote account, or subscription shall be required for any function of the unit.

3.4.5 Operator hardware controls shall not be subject to automated lockout.

3.4.6 Operator hardware controls shall be sized and spaced for actuation by a gloved hand. The reference case is a structural firefighting glove; where a lighter wildland or extrication glove is the design basis, the installation documentation shall state the glove class assumed, and control sizing shall follow from it.

3.5 Antennas and Positioning

3.5.1 Radio antennas shall be external via weatherproof RF bulkheads or assembled penetrations.

3.5.2 Each unit shall carry a GNSS receiver. The antenna shall be placed with open sky view: external active antenna via bulkhead, external sealed puck on data cable, or integrated through-panel unit per 3.1.3.

3.6 Marking and Labeling

3.6.1 Each unit shall bear a permanent label identifying: specification and annex version built to; supply voltage; battery type and replacement rating; and the drain-down location.

3.6.2 The notification button shall be labeled with words to the effect that it signals participating systems and is not a substitute for contacting emergency services.

3.6.3 Zone outlets shall be individually and permanently identified.

3.7 Accessory Water Appliances

3.7.1 Coordination-capable water appliances (remotely actuated valves, gates, wyes, manifolds, and supply-point assemblies) are accessories, not required equipment. A compliant installation of any class requires none of them.

3.7.2 All wetted appliance connections shall use national hose thread (NH/NST) or NPT as appropriate to the connection point. Proprietary couplings shall not be used.

3.7.3 Appliances attaching to hydrants, standpipes, or utility fittings shall be clamp-on or thread-on, shall not permanently modify the fitting, and their removal shall restore the fitting to its original condition and operation.

3.7.4 Every actuated appliance shall preserve manual operation. Failure or de-energization of the actuator shall leave the appliance manually operable by standard tools or standard practice.

3.7.5 Battery-powered appliances shall use a major open tool-platform battery interface per 6.5. Appliances requiring proprietary single-source battery systems shall not be permitted.

3.7.6 Appliance installation on utility infrastructure (hydrants, district valves) shall be subject to the utility’s or district’s approval and standards; nothing in this document waives that authority.

Chapter 4 — Class 1: Residential

4.1 The enclosure shall be NEMA 4X, polycarbonate or stainless. The reference size for a residential node is nominal 16 in. × 20 in. × 8 in.; larger enclosures are compliant provided they house the specified internal panel layout and preserve the wet/dry separation of §3.

4.2 Penetrations shall consist of: one 3/4 in. NPT inlet; three 3/4 in. NPT outlets; one N-type RF bulkhead (top); one SMA RF bulkhead (top); and one cord-grip strain relief.

4.3 Power shall be 120 VAC cord-and-plug through the strain relief.

4.4 Batteries shall be two 12 V, 7–9 Ah sealed lead-acid units in series, or a drop-in 24 V LiFePO4 equivalent.

4.5 Zone valves shall be standalone motorized full-port brass ball valves with actuator pads and position feedback, fed by the manifold header per 3.3.6, one per independently commanded zone. *The reference residential node uses three 3/4 in. valves at 24 VDC; zone count and valve size follow from the supply and coverage plan of the installation.* Solenoid valves are a permitted budget variant where position feedback is deliberately waived.

4.6 External connections shall use commodity means: cord grips, DIN terminal blocks, NPT/GHT. Keyed or proprietary connectors shall not be used.

4.7 The unit shall self-calibrate to available supply by observing inlet pressure under single-zone flow. No hydraulic configuration by the installer or owner shall be required.

4.8 Default operation shall be sequencing per 2.4. Simultaneous zone operation shall not be assumed in design or represented in installation documents.

Chapter 5 — Class 2: Government / Commercial

5.1 Class 2 units shall meet all Chapter 3 and Chapter 4 requirements except as modified by this chapter.

5.2 External connections shall use sealed circular connectors of an open multi-vendor standard; M12 is the reference class. Deutsch DT and MIL-DTL-38999 are permitted where procurement requires.

5.3 For marine and coastal exposure, wetted fittings shall be 316 stainless steel and polymer enclosures are preferred.

5.4 GNSS shall be an integrated through-panel marine-grade unit per 3.1.3 as standard.

5.5 An internal DIN-rail unmanaged Ethernet switch, minimum 5 ports, 24 VDC, shall be standard equipment. Managed switches are permitted where agency network policy requires; the reference installation shall not require network configuration to commission.

5.6 Power shall be hardwired or cord-and-plug per site requirements, on the same DIN/commodity battery architecture as Class 1.

5.7 Agency-furnished radio and transport equipment shall integrate via standard Ethernet interfaces and shall be installed per the furnishing agency’s requirements. Units shall not transmit on fire voice networks; agency radio equipment operates under the furnishing agency’s authority and applicable interagency listings.

Chapter 6 — Class 3: Tactical / Transportable

6.1 The enclosure shall be a ruggedized transit case. Interior components are not required to be individually sealed; the case provides the environmental boundary in transport and the open lid is the service interface in operation.

6.2 The case, as shipped ready to commission, shall not exceed 40 lb (18 kg). Kit components carried separately (power station, solar panel, gateway terminal) shall each individually not exceed the same limit.

6.3 Every case shall provide handles sized and positioned for both one-person and two-person carry.

6.4 Primary power shall be battery/solar: a portable power station of an open commodity class or internal LiFePO4 with charge controller.

6.5 An auxiliary power dock shall be provided accepting a major open tool-platform battery pack, selected at order to match the operating department’s existing battery fleet. Equipment requiring proprietary single-source battery systems shall not be permitted at this interface.

6.6 One or more interior USB Power Delivery outlets, minimum 45 W each, shall be provided. Outlets shall deliver rated power whenever energy is physically present; automated policy shall inform but shall not refuse. A charging cable of the prevailing consumer interface shall ship in the case and shall be inventoried as a consumable.

6.7 Hydraulic sensing shall be by an inline sensor spool with 1.5 in. NH forestry couplings. Engine sensing shall be by non-contact acoustic means. No baseline sensor shall touch, mount to, or modify pump equipment.

6.8 Units shall commission by voice-guided setup: each instruction shall gate on a sensor-verified condition, and the unit shall announce readiness on completion.

6.9 Prepositioned units shall require no external activation, registration, provisioning service, or connectivity to enter service. Warehouse maintenance shall be limited to battery inspection/replacement and consumable replacement on standard storage intervals.

6.10 A satellite emergency module (dual-constellation) is permitted as the expected configuration; the unit’s fire defense function and local distress signaling shall not depend on satellite service.

Chapter 7 — Documentation

7.1 Each installation shall be documented with: a written narrative of system intent and coverage; a zone layout identifying coverage areas; the water supply type and measured or rated delivery; and the electrical supply arrangement.

7.2 Completion documentation shall record the self-calibration result (supply behavior and cycle timing as set) and a functional test of each zone valve, each hardware control, and the audible annunciation.

7.3 The unit’s internal event log shall be regarded as the operational record. The installer shall verify at completion

that the log is recording.

7.4 A copy of the installation documentation shall remain with the unit (interior document holder) and with the owner.

End of document. All requirements above derive from the Hardware Annex and Main Specification; discrepancies found in review supersede nothing automatically but will be dispositioned against the source documents. This draft is structured for direct use as public input to relevant standards-development processes.